

M. Sc. Thesis' Abstract

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Abstract

In Probability Theory, there is an area that studies spectral properties of large matrices with random inputs. One of the most important is Wigner's Semicircle Law, which opens a broad theory called Random Matrix Theory. This research replicates some results of this theory and explores its asymptotic, numerical, and practical extents.

In particular, this thesis presents an elementary demonstration of the Local Semicircle Law. This law is a refinement of Wigner's law when combined with the inverse Stieltjes transform, another limit result. This combination serves to bound the approximation error. This proof is intended to serve as a first introduction to the topic of local laws, with the declared intention to skip the specific notation of 'Stochastic Domination' commonly used in this theory. For the sake of simplicity, the proof rests on an elementary concentration inequality for Gaussian vectors.

The ideas of this proof are extended to present results in the literature on sparse matrices. In this thesis, a recent result is approached from a numerical point of view to exemplify and complement the theoretical results.

Keywords: *random matrices, Wigner's semicircle law, local laws, local semicircle law, GUE, concentration inequalities, large deviations, sparse matrices*